

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 34)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Give **three** sources of oil pollution of the sea other than tanker accidents.

[3 marks]

1

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2

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3

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0 1 . 2

Explain how an oil spill at sea can kill seabirds.

[3 marks]

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0 1 . 4

Explain why oil pollution is most serious when it occurs inshore or in an enclosed body of water.

[2 marks]

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10

0 2 Table 1 compares two types of satellite used to monitor the Earth.

Table 1

Feature	Low Earth Orbit (polar) satellite	Geostationary satellite
Altitude	About 800 km	36 000 km above the equator
Orbit period	About 1.5 hours	24 hours (fixed position)
Coverage	Whole Earth surveyed in about 15 days	Same third of the Earth continuously
Detail	High resolution	Lower resolution

0 2 . 1 Use Table 1 to explain why LEO satellites are used to monitor changes in ice cover and vegetation, whereas geostationary satellites are used to monitor weather.

[3 marks]

0 2 . 2 Give **three** climate-related factors, other than ice cover, that are measured by satellite sensors.

[3 marks]

1

2

3

02 . 3

Explain how a computer climate model can be tested for reliability before its predictions are used.

[2 marks]

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02 . 4

Explain why representative data are essential for climate modelling.

[2 marks]

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10

03 . 1

Explain how each of the following building design features reduces energy use.

[6 marks]

Orientation

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Low surface area : volume ratio

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Earth sheltering

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03 . 2

Compare the embodied energy of concrete with alternative building materials.

[2 marks]

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03 . 3

Describe how a heat exchanger allows a building to be ventilated with minimal heat loss.

[2 marks]

10

04 . 1

Give **three** sources of mercury pollution.

[3 marks]

1

2

3

04 . 2

Describe **three** methods used to control mercury pollution.

[3 marks]

1

2

3

0 4 . 3

Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

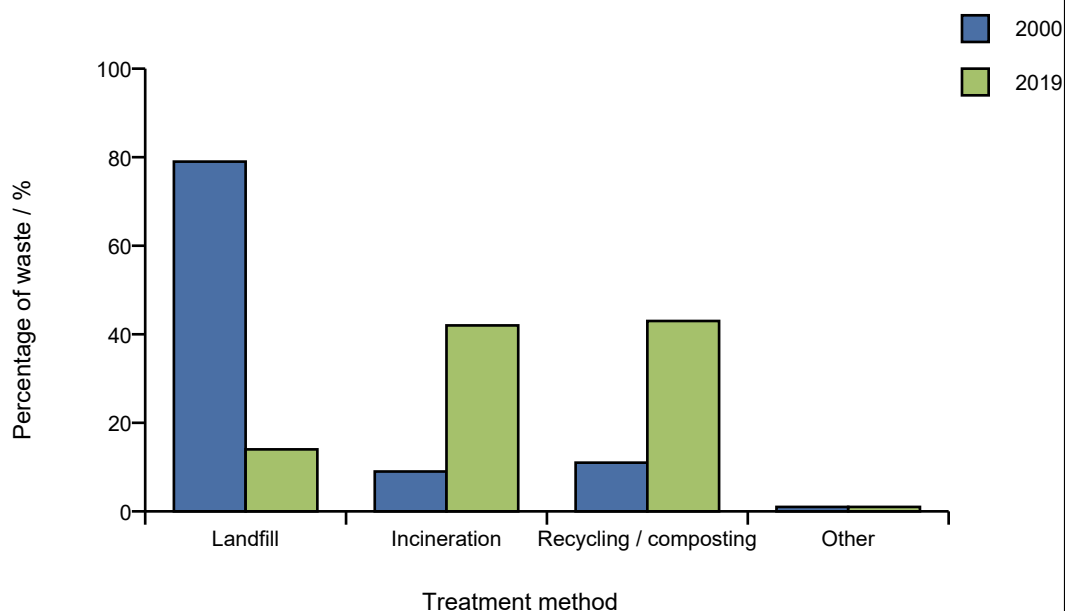
0 4 . 4

State what is meant by synergism and give an example involving a heavy metal.

[2 marks]**10**

0 5

Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1

0 5 . 1

Describe the changes shown in Figure 1.

[2 marks]

05 . 2

Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

1

2

3

05 . 3

Describe **four** features of a well-managed landfill site.

[4 marks]

1

2

3

4

05 . 4

Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

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10

06

A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

06

. 1

Calculate the estimated population size.

[2 marks]

Answer

beetles

06

. 2

State **four** assumptions of the Lincoln Index method.

[4 marks]

1

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2

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3

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4

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06

. 3

Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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06

. 4

Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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10

0 7 Table 1 shows the water quality required for four different uses.

0 7 . 1 Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

0 7 . 2 Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

07 . 3

Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

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07 . 4

Give **two** methods used to sterilise treated water.

[2 marks]

1

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2

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10

08 . 1

Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

- A High sand content
- B High clay content
- C Crumb peds
- D High pH

☐
☐
☐
☐

08 . 2

Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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08 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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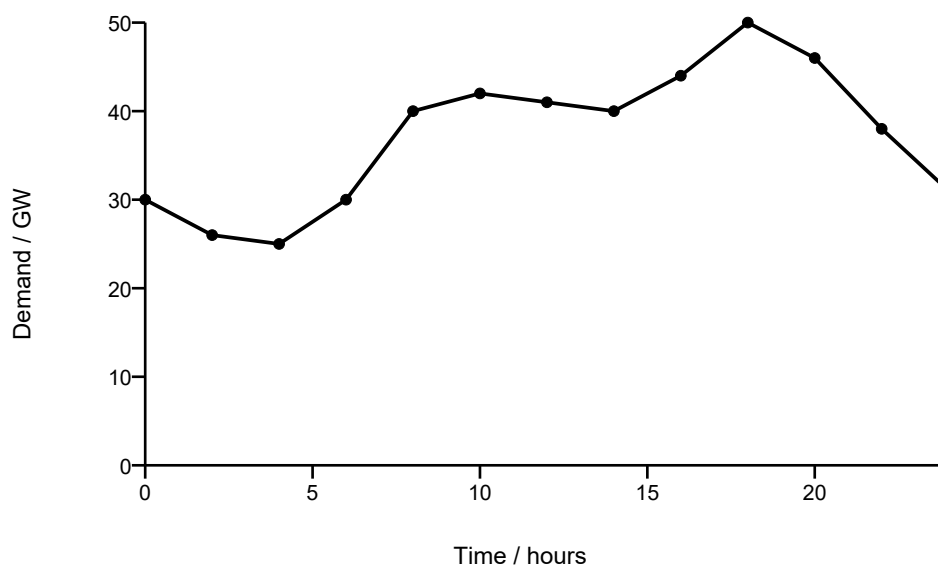
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5

09 . 9

Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1



09 . 1

Describe the pattern of demand shown in Figure 1.

[2 marks]

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09 . 2

Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

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09 . 3

Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

09 . 4

Give **two** causes of fluctuations in energy **supply**.

[2 marks]

1

2

10

10 . 1

Evaluate the potential contribution of renewable energy resources to future energy supplies in a country such as the UK.

[9 marks]

[illegible]

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10 . 2

State the factor by which the kinetic energy of wind increases when wind velocity doubles.

[1 mark]

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10

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Write an essay on one of the following topics.

1	1	.	1
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Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS